3.1 Control

2-D Quadrotor Control

$$\begin{bmatrix} \dot{y} \\ \dot{z} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{m}\sin\phi & 0 \\ \frac{1}{m}\cos\phi & 0 \\ 0 & \frac{1}{I_{xx}} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$x = \begin{bmatrix} y \\ z \\ \phi \end{bmatrix} \quad \dot{x} = \begin{bmatrix} \dot{y} \\ \dot{z} \\ \dot{\phi} \\ \ddot{y} \\ \ddot{z} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} 0 \\ -g \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\frac{1}{m}\sin\phi & 0 \\ \frac{1}{m}\cos\phi & 0 \\ 0 & \frac{1}{I_{xx}} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Equations of motion

$\ddot{y} = -\frac{1}{m}u_1\sin(\phi)$ 悬停时 $\phi = 0$, $u_1 = mg$, $u_2 = 0$

$\ddot{z} = -g + \frac{1}{m}u_1\cos(\phi)$

$\ddot{\phi} = \frac{u_2}{I_{xx}}$ Nonlinear

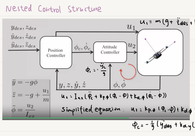
Trajectory Tracking

$r_d(t) = \begin{bmatrix} y(t) \\ z(t) \end{bmatrix}$

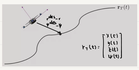
$e_p = r_d(t) - r$ want

$e_v = \dot{r}_d(t) - \dot{r}$ $(\ddot{r}^d(t) - \ddot{r}_c) + k_d e_v + k_p e_p = 0$

Nested Control Structure



3-D Quadrotor Control



$e_p = \ddot{r}_d(t) - r$

$e_v = \dot{r}_d(t) - \dot{r}$

want $(\ddot{r}_d(t) - \ddot{r}_c) + k_d e_v + k_p e_p = 0$

Commanded acceleration, calculated by the controller

$$u_2 = \begin{bmatrix} k_{p,\phi}(\phi_c - \phi) + k_{d,\phi}(p_c - p) \\ k_{p,\theta}(\theta_c - \theta) + k_{d,\theta}(q_c - q) \\ k_{p,\psi}(\psi_c - \psi) + k_{d,\psi}(r_c - r) \end{bmatrix}$$

$$(\ddot{r}_{i,des} - \ddot{r}_{i,c}) + k_{d,i}(\dot{r}_{i,des} - \dot{r}_c) + k_{p,i}(r_{i,des} - r_c) = 0$$

actual (feedback)

specified

command $u_1 = m$

Control of Hovering 悬停状态下, 通过和零力矩相抵, 所有速度为0 roll and pitch = 0

$m\ddot{r} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + R \begin{bmatrix} 0 \\ 0 \\ F_1 + F_2 + F_3 + F_4 \end{bmatrix}$ Linearization $\sin\phi \approx \phi$, $\cos\phi \approx 1$ $\forall \phi = 0$

$r_i = \ddot{x}, y(\cos\psi + \phi\sin\psi)$

$\ddot{r}_i = \ddot{x}, y(\sin\psi - \phi\cos\psi)$

$u_1 = m(g + \ddot{r}_z)$

$I \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \underbrace{\begin{bmatrix} L(F_3 - F_4) \\ L(F_2 - F_1) \\ M_1 - M_2 + M_3 - M_4 \end{bmatrix}}_{u_2 = 0} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times I \begin{bmatrix} p \\ q \\ r \end{bmatrix}$ 悬浮状态下, 通过和零力矩相抵, 所有速度为0

$$\boxed{\begin{aligned} \phi_c &= \frac{1}{g}(\ddot{r}_{x,c}\sin\psi_{des} - \ddot{r}_{y,c}\cos\psi_{des}) \\ \theta_c &= \frac{1}{g}(\ddot{r}_{x,c}\cos\psi_{des} - \ddot{r}_{y,c}\sin\psi_{des}) \end{aligned}}$$

$\psi_c = \psi^{des}$

3.2 Planning

Time, Motion, and Trajectories

General Set Up

- Start, goal positions (orientations)
- Waypoint positions (orientations)
- Smooth criterion

  Generally translates to minimizing rate of change of "input".
- Order of the system (n)

  Order of the system determines the input

  Boundary conditions on $(n-1)^{th}$ order and lower derivatives (微分)

Calculus of Variations

$x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T \mathcal{L}(\dot{x}, x, t)\,dt$

Examples:
- Shortest distance path (geometry) $x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T \dot{x}^2\,dt$ 最短路径
- Fermat's principle (optics) $x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T 1\,dt$ 最短时间
- Principle of least action (mechanics) $x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T \underbrace{T(\dot{x}, x, t)}_{\text{动能}} - \underbrace{V(x, t)}_{\text{势能}}\,dt$

Euler Lagrange Equation:

1阶变分, 欧拉-拉格朗日方程 $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0$

Smooth trajectories (n=1)

$x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T \dot{x}^2\,dt$ Solution $\mathcal{L}(\dot{x}, x, t) = \dot{x}^2$ $\ddot{x} = 0$

$x(0) = x_0$ $x(T) = x_T$ $\longrightarrow$ $x = C_1 t + C_0$

Smooth trajectories (general n)

$x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T (x^{(n)})^2\,dt$ n阶导数 $u = x^{(n)}$

Euler-Lagrange Equation:

1阶变分, $\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) + \frac{d^2}{dt^2}\left(\frac{\partial \mathcal{L}}{\partial \ddot{x}}\right) + \dots + (-1)^n \frac{d^n}{dt^n}\left(\frac{\partial \mathcal{L}}{\partial x^{(n)}}\right) = 0$

n=1, shortest distance

n=2, min acceleration

n=3, min jerk

n=4, min snap

Example: $x(0) = a$ $x(T) = b$

$x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T \mathcal{L}(\ddot{x}, \dot{x}, x, t)\,dt$

$\mathcal{L} = (\ddot{x})^2$

Euler-Lagrange: $\boxed{\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) + \frac{d^2}{dt^2}\left(\frac{\partial \mathcal{L}}{\partial \ddot{x}}\right)} - \frac{d^3}{dt^3}\left(\frac{\partial \mathcal{L}}{\partial \dddot{x}}\right) = 0$ $= 0$

$x^{(4)} = 0$

$x = C_4 t^3 + C_3 t^2 + C_2 t^1 + C_1 t^0 + C_0$

Boundary conditions:

$\dot{x} = 3C_4 t^2 + 2C_3 t^1 + C_2$

$\ddot{x} = 6C_4 t + 2C_3$

$\dddot{x} = 6C_4$

	Position	Velocity	Acceleration
$t=0$	a	0	0
$t=T$	b	0	0

$$\begin{bmatrix} a \\ b \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ T^5 & T^4 & T^3 & T^2 & T^1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5T^4 & 4T^3 & 3T^2 & 2T & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 20T^3 & 12T^2 & 6T & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} C_5 \\ C_4 \\ C_3 \\ C_2 \\ C_1 \\ C_0 \end{bmatrix}$$

argmin$_{x(t)} \int_0^T \|x^{(k)}(t)\|^2 dt$ is an kth degree polynomial trajectory where k = 4

$x^{(k)}(t) = C_k x^k + C_{k-1} x^{k-1} + C_{k-2} x^{k-2} + \dots + C_1 x + C_0$

手动写所有数方程, 然后写为 4

简化解决
- 近似手动写出方程
- 角速度和速度平滑
- roll and pitch 都为 0

Extensions to multiple dimensions

$(x^*(t), y^*(t)) = \underset{x(t), y(t)}{\operatorname{argmin}} \int_0^T \mathcal{L}(\dot{x}, \dot{y}, x, y, t)\,dt$

Euler Lagrange Equation: $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0$

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}}\right) - \frac{\partial \mathcal{L}}{\partial y} = 0$

Multi-Segment 1D Trajectories

Design a trajectory $x(t)$ such that: $t = [t_0\ t_1\ t_2 \dots t_m]^T$

$x = [x_0\ x_1\ x_2 \dots x_m]^T$

= 阶梯函数 $\underset{x(t)}{\min} \int_{t_0}^{t_m} (\ddot{x}(t))^2\,dt + \dots + \int_{t_{m-1}}^{t_m} (\ddot{x}(t))^2\,dt$

Cubic Spline

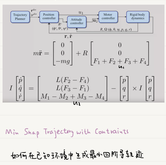
$$x(t) = \begin{cases} x_1(t) = C_{1,3}t^3 + C_{1,2}t^2 + C_{1,1}t + C_{1,0}, & t_0 \le t \le t_1 \\ x_2(t) = C_{2,3}t^3 + C_{2,2}t^2 + C_{2,1}t + C_{2,0}, & t_1 \le t \le t_2 \\ \dots \\ x_m(t) = C_{m,3}t^3 + C_{m,2}t^2 + C_{m,1}t + C_{m,0}, & t_{m-1} \le t \le t_m \end{cases}$$

三次样条

4n degrees of freedom

这里的目标是最小的轨迹精度, 此时设计问题

Motion Planning for Quadrotors



The position control system is a fourth order system

want trajectories that can be differentiated four times

Minimum Snap Trajectory

$x^*(t) = \underset{x(t)}{\operatorname{argmin}} \int_0^T (x^{(4)})^2\,dt$

最小四阶导轨迹

Min Snap Trajectory with Constraints

如何在之间轨迹中主要最小四阶导轨迹

Obstacles

$\vec{r}_{obs} - \vec{r}(t) \in S_{obs}$

Integer Constraints for Obstacle Avoidance

$\vec{r}_{obs} - \vec{r}(t_i) \in S_{obs} \to \text{Mixed}$ $\forall t = t_1, \dots, t_N(t)$

悬浮状态, 避免人类碰撞

最短速度曲线也是最短距离曲线